MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 10

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Recap of Previous Lecture

Continuous-Time Dynamics

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\boldsymbol{\psi}_t(\mathbf{x}_0) = \int_{t_0}^{t} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(s), s) ds + \mathbf{x}_0 \approx \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0, t).$$

- $\mathbf{v}_{\boldsymbol{\theta}} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ is a vector field.
- $\boldsymbol{\psi} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ is a flow (the solution of ODE):

Euler `ODESolve`

$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

For a smooth field on a fixed interval, compare global error and **NFE** (velocity-network evaluations):

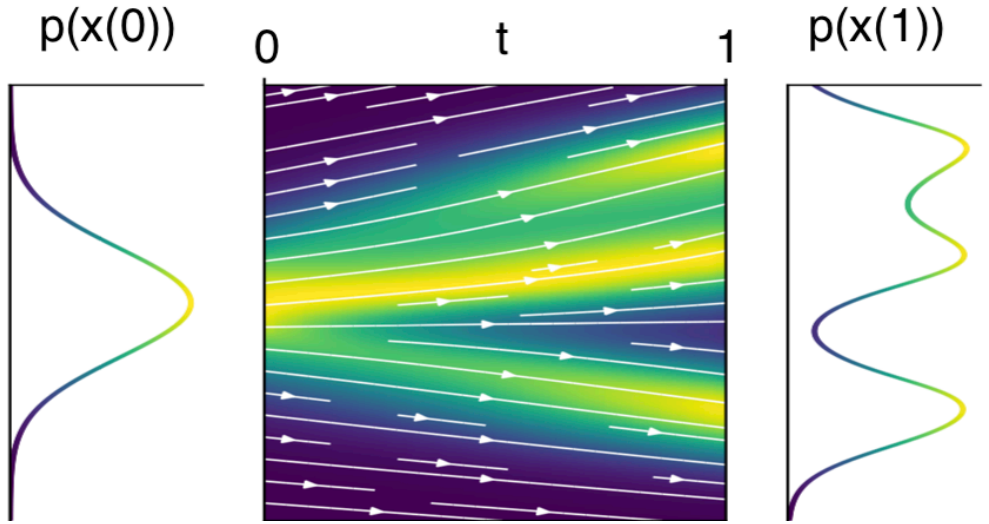
- **Euler:** $\mathcal{O}(h)$ global error, **1** evaluation per step.
- **Heun:** $\mathcal{O}(h^2)$ global error, **2** evaluations per step, averaging the current and predicted endpoint velocities.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Driscoll T. A., Braun R. J. Fundamentals of Numerical Computation: Runge–Kutta methods

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

- Suppose $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$. Then, $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.
- $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.



Recap of Previous Lecture

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t, then the ODE has a **unique solution** given by a flow $\boldsymbol{\psi}_t$.

This guarantees the ODE is **uniquely reversible**.

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\, dt$$

Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in t, and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018

Recap of Previous Lecture

Fix $p_0 = \mathcal{N}(0, \mathbf{I})$ and learn $\mathbf{v}_{\boldsymbol{\theta}}$; the endpoint density is $p_1 = p_{\boldsymbol{\theta}}$.

Training

1. Sample $\mathbf{x} \sim p_{\text{data}}$. Solve the ODE **backward**, from $\mathbf{x}(1) = \mathbf{x}$ to $\mathbf{x}(0)$.
2. Compute the negative log-likelihood along this trajectory:

$$\mathcal{L} = -\log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\log p_0(\mathbf{x}(0)) + \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

3. Update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}$; differentiate through the ODE, e.g., via the **adjoint method**.

Sampling

1. Sample $\mathbf{x}(0) \sim p_0$.
2. Solve the ODE **forward** and return the endpoint:

$$\mathbf{x}(1) = \texttt{ODESolve}_v(\mathbf{x}(0), \boldsymbol{\theta}, t_0 = 0, t_1 = 1) \sim p_{\boldsymbol{\theta}}.$$

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

Recap of Previous Lecture

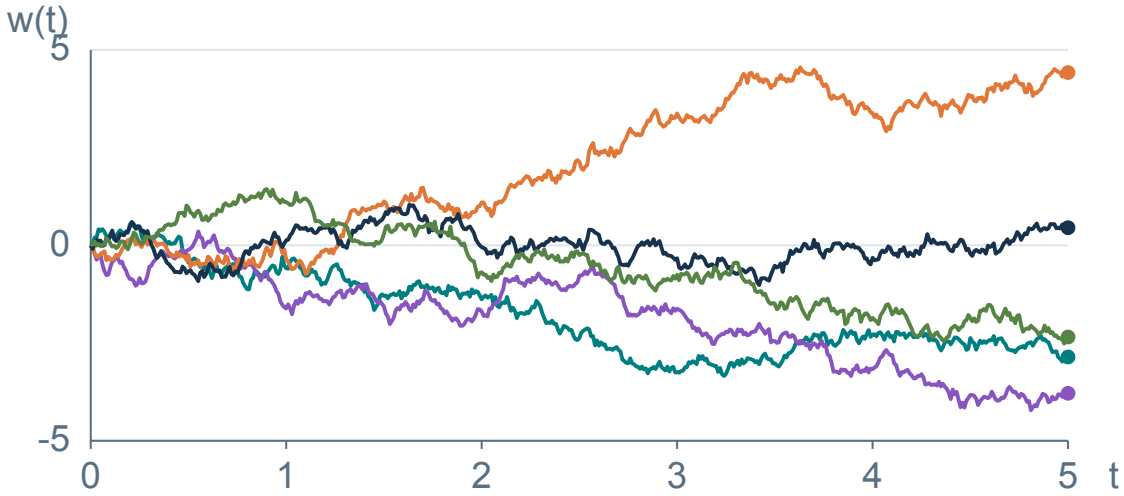
SDE Basics

An SDE generalizes the dynamics of a CNF by adding Brownian noise to the motion induced by the velocity field.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}, \qquad \mathbf{x}(0) \sim p_0.$$

Wiener Process

$$\Delta\mathbf{w} = \mathbf{w}(t + \Delta t) - \mathbf{w}(t) \sim \mathcal{N}(0, \Delta t\, \mathbf{I}), \qquad \Delta t > 0.$$



Here $d\mathbf{w}$ is **informal shorthand** for a small Brownian increment, modeled as $\sqrt{dt}\,\boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

Discretizing the SDE (Euler-Maruyama Update) – `SDESolve`

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \boldsymbol{\epsilon} \cdot \sqrt{dt}$$

- At each time t, we have the density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- $p : \mathbb{R}^m \times [0, 1] \to \mathbb{R}_+$ is a **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.

Recap of Previous Lecture

Two Sources of Randomness

Randomness can enter an SDE through the **initial state** $\mathbf{x}(0) \sim p_0$ and the **Brownian path** $\mathbf{w}(t)$.

Theorem (Kolmogorov-Fokker-Planck)

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., C^1 in t and C^2 in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})\right) + \frac{1}{2} g^2(t) \Delta_{\mathbf{x}} p_t(\mathbf{x})$$

Langevin SDE (Special Case)

$$d\mathbf{x} = \tfrac{1}{2} \nabla_{\mathbf{x}} \log p_0(\mathbf{x})\, dt + d\mathbf{w}.$$

If $\mathbf{x}(0) \sim p_0$, then $\mathbf{x}(t) \sim p_0$ for every t.

Set the target density to the model density, $p_0(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$.

Langevin Dynamics

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{\eta}{2} \nabla_{\mathbf{x}_k} \log p_{\boldsymbol{\theta}}(\mathbf{x}_k) + \sqrt{\eta}\, \boldsymbol{\epsilon}_k, \qquad \boldsymbol{\epsilon}_k \sim \mathcal{N}(0, \mathbf{I}).$$

Holderrieth P., Erives E. An Introduction to Flow Matching and Diffusion Models, 2025, §3.3
Welling M. Bayesian Learning via Stochastic Gradient Langevin Dynamics, 2011

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

Outline

01 **Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)**

02 Probability Flow ODE

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Score Matching SDE

Two consecutive noise levels

$$\mathbf{x}_{k-1} = \mathbf{x} + \sigma_{k-1}\boldsymbol{\epsilon}_{k-1}$$
$$\mathbf{x}_k = \mathbf{x} + \sigma_k\boldsymbol{\epsilon}_k$$
$$q(\mathbf{x}_j \mid \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_j^2\mathbf{I}), \quad j \in \{k-1, k\}.$$

Markov step from $k-1$ to k

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \sqrt{\sigma_k^2 - \sigma_{k-1}^2}\,\boldsymbol{\epsilon}$$
$$q(\mathbf{x}_k \mid \mathbf{x}_{k-1}) = \mathcal{N}\big(\mathbf{x}_{k-1}, (\sigma_k^2 - \sigma_{k-1}^2)\mathbf{I}\big)$$

From a finite step to continuous time

$$\mathbf{x}(t) - \mathbf{x}(t-\Delta t) = \sqrt{\sigma^2(t) - \sigma^2(t-\Delta t)}\,\boldsymbol{\epsilon}$$
$$= \sqrt{\frac{\sigma^2(t) - \sigma^2(t-\Delta t)}{\Delta t}}\,\underbrace{\sqrt{\Delta t}\,\boldsymbol{\epsilon}}_{\Delta\mathbf{w}}$$

As $\Delta t \to 0$, the variance rate sets the diffusion coefficient:

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}}\,d\mathbf{w}.$$

Variance Exploding SDE (VE-SDE)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

Drift and diffusion

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

Why the variance grows

$\sigma(t)$ is a monotonically increasing function. From time s to t, independent noise increments add variance in each coordinate:

$$\int_s^t g^2(u)\, du = \sigma^2(t) - \sigma^2(s).$$

Diffusion SDE

One DDPM step

$$\mathbf{x}_k = \sqrt{1-\beta_k}\,\mathbf{x}_{k-1} + \sqrt{\beta_k}\,\boldsymbol{\epsilon}_k, \qquad q(\mathbf{x}_k \mid \mathbf{x}_{k-1}) = \mathcal{N}\left(\sqrt{1-\beta_k}\,\mathbf{x}_{k-1},\ \beta_k \mathbf{I}\right).$$

From variance per step to variance per unit time

Split $[0,1]$ into T steps of length $\Delta t = 1/T$. Step k ends at $t_k = k/T$.

$$\underbrace{\beta_k}_{\text{noise variance per step}} = \underbrace{\beta(t_k)}_{\text{noise variance rate}} \cdot \underbrace{\Delta t}_{\text{step length}}.$$

At a constant rate, doubling T halves the noise variance added at each step.

Substitute into the update at $t = t_k$

$$\mathbf{x}(t) = \sqrt{1-\beta(t)\Delta t}\,\mathbf{x}(t-\Delta t) + \sqrt{\beta(t)\Delta t}\,\boldsymbol{\epsilon}_k$$

$$\mathbf{x}(t) - \mathbf{x}(t-\Delta t) \approx -\tfrac{1}{2}\beta(t)\mathbf{x}(t-\Delta t)\,\Delta t + \sqrt{\beta(t)}\,\underbrace{\sqrt{\Delta t}\,\boldsymbol{\epsilon}_k}_{\Delta \mathbf{w}}.$$

Variance Preserving SDE (VP-SDE)

DDPM increment at $t = t_k$

$\mathbf{x}(t) - \mathbf{x}(t - \Delta t) \approx -\frac{1}{2}\beta(t)\mathbf{x}(t - \Delta t)\,\Delta t + \sqrt{\beta(t)}\,\Delta\mathbf{w}.$

Here $\Delta t = 1/T$ and $\Delta\mathbf{w} = \mathbf{w}(t) - \mathbf{w}(t - \Delta t) = \sqrt{\Delta t}\,\boldsymbol{\epsilon}_k.$

As $\Delta t \to 0$, the DDPM update gives the continuous-time dynamics

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

From Score Matching and Gaussian Diffusion to SDEs

Score matching → NCSN

Fit one denoising score network across noise levels.

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}.$$

Gaussian diffusion → DDPM

Predict the added noise to learn the reverse chain.

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}.$$

Weighted score matching → noise-prediction MSE. Let r_t be the noise standard deviation.

$$r_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \boldsymbol{\epsilon}/r_t \right\|^2 = \left\| \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \boldsymbol{\epsilon} \right\|^2, \qquad \boldsymbol{\epsilon}_{\boldsymbol{\theta},t} = -r_t \mathbf{s}_{\boldsymbol{\theta},t}.$$

ODE → SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt \xrightarrow{\text{add Brownian noise}} d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}.$$

Variance Exploding SDE (NCSN)

This process matches the NCSN noise levels.

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}}\, d\mathbf{w}.$$

Variance Preserving SDE (DDPM)

This is the small-step limit of the DDPM forward chain.

$$d\mathbf{x} = -\tfrac{1}{2}\beta(t)\mathbf{x}(t)\, dt + \sqrt{\beta(t)}\, d\mathbf{w}.$$

Song & Ermon, Generative Modeling by Estimating Gradients of the Data Distribution, 2019 · Ho et al., Denoising Diffusion Probabilistic Models, 2020
Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Outline

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From an SDE to a Probability Flow ODE

Neural ODE and continuity equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt \quad \Longrightarrow \quad \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) \quad \Longleftrightarrow \quad \frac{\partial p_t}{\partial t} = -\operatorname{div}(p_t \mathbf{v})\,.$$

SDE and Fokker–Planck equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w} \qquad \Longrightarrow \qquad \frac{\partial p_t}{\partial t} = -\operatorname{div}(p_t \mathbf{f}) + \tfrac{1}{2} g^2(t) \Delta_{\mathbf{x}} p_t.$$

Sources of randomness

1. **Initial state** $\mathbf{x}(0) \sim p_0$ (ODE and SDE). This draw determines the entire ODE trajectory.
2. **Wiener process** $\mathbf{w}(t)$ (SDE only). Setting $g(t) = 0$ removes this noise and recovers the ODE with $\mathbf{v} = \mathbf{f}$.

With diffusion, match the probability flux

$$\underbrace{p_t \mathbf{v}}_{\text{ODE flux}} = \underbrace{p_t \mathbf{f} - \tfrac{1}{2} g^2(t) \nabla_{\mathbf{x}} p_t}_{\text{SDE flux}}\,.$$

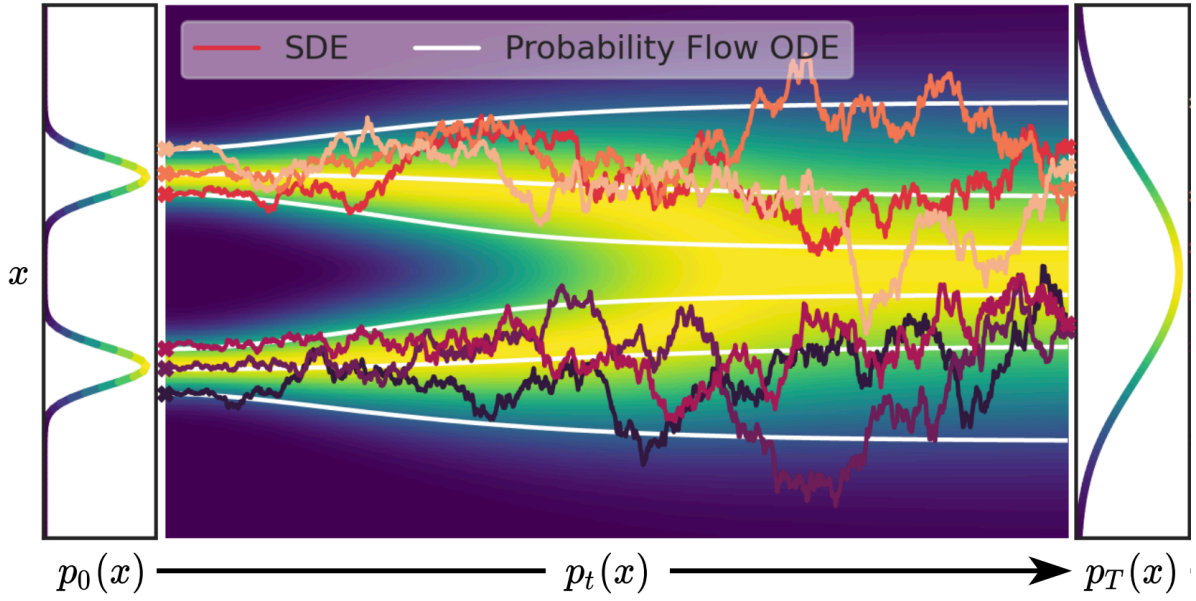
From the same p_0, this ODE gives the **same density p_t at every time t**. Individual trajectories can differ.

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$



Probability Flow ODE

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Proof

$$\begin{aligned}
\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right]\right) \\
&= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}}\log p_t(\mathbf{x})\right)p_t(\mathbf{x})\right]\right)
\end{aligned}$$

Probability Flow ODE

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) p_t(\mathbf{x})\right]\right) \\ &= \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right]\right) = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right)\end{aligned}$$

This is the continuity equation for the velocity field $\mathbf{v}$.

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}), \qquad \tilde{g}(t) = 0.$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt + 0 \cdot d\mathbf{w} = \left(\mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt.$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})) + \underbrace{\tfrac{1}{2} \tilde{g}^2(t) \Delta_{\mathbf{x}} p_t(\mathbf{x})}_{0}.$$

> Starting from the same p_0, the ODE and SDE have the same density p_t at every time.

Probability Flow ODE

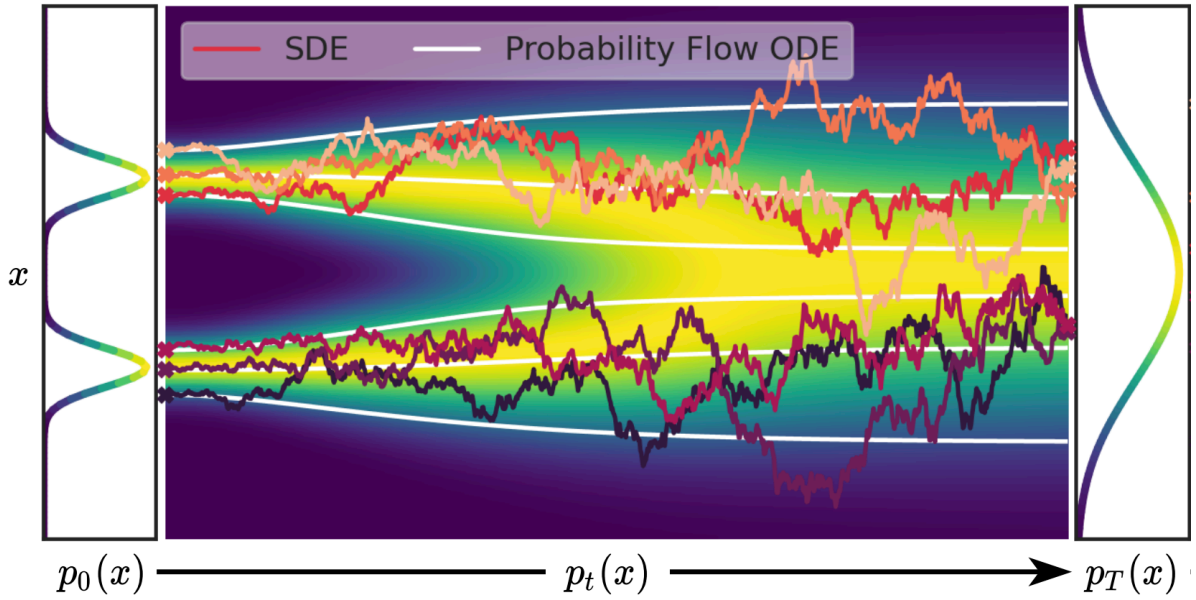
SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}.$$

Probability Flow ODE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt = \left[\mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] dt.$$

The **continuous-time score** $\mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ determines the correction to the drift.



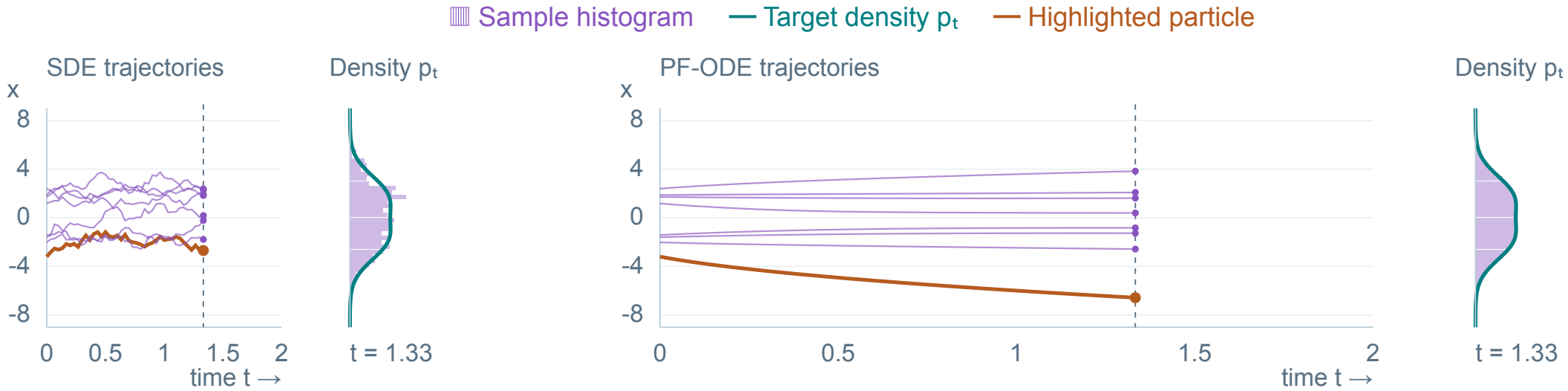
The white curves show the smooth ODE trajectories. The colored curves show SDE trajectories.

Same Densities, Different Trajectories

Both dynamics have density $p_t(x) = \frac{1}{2}\mathcal{N}(x \mid -a, v_t) + \frac{1}{2}\mathcal{N}(x \mid a, v_t)$, with $a = 1.8$ and $v_t = 0.25 + 2t$.

$$dx = \sqrt{2}\, dw, \qquad dx = -s(x,t)\, dt, \qquad s(x,t) = \partial_x \log p_t(x).$$

Same initial samples, shown from t = 0 to t = 1.33.



The same density at every time does not require the same particle trajectories.

Why use a probability flow ODE?

Deterministic sampling

Each draw $\mathbf{x}(T) \sim p_T$ determines a unique sample $\mathbf{x}(0)$. Solve backward from T to 0 using

$$d\mathbf{x} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\, dt, \qquad \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2} g^2(t) \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t).$$

Choice of solver

- Euler and Heun use one and two network evaluations per step, respectively.
- Other solvers include **DPM-Solver**, **DPM-Solver++**, and **UniPC**.

Compare sampling cost by the total **number of network evaluations (NFE)**.

Log-likelihood

The flow is invertible. Follow the same ODE from data to noise to evaluate the model density.

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}(0)) = \log p_T(\mathbf{x}(T)) + \int_0^T \operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

Song et al., Score-Based Generative Modeling through SDEs, 2020 · Chen et al., Neural Ordinary Differential Equations, 2018
Karras et al., Elucidating the Design Space of Diffusion-Based Generative Models, 2022 · Lu et al., DPM-Solver, 2022 · Lu et al., DPM-Solver++, 2022 · Zhao et al., UniPC, 2023

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

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How do we take one step backward?

Given the state now, where was the particle a short time h earlier?

ODE · One predecessor

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt$$



Over a short forward step h,

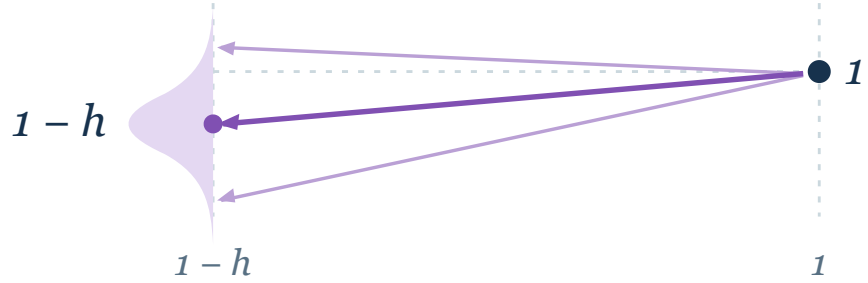
$$\mathbf{x}(t+h) \approx \mathbf{x}(t) + h\,\mathbf{v}(\mathbf{x}(t), t).$$

In reverse time $\tau = 1 - t$,

$$d\mathbf{x} = -\mathbf{v}(\mathbf{x}, 1-\tau)\, d\tau.$$

SDE · A Wiener process

Start at $w(0) = 0$ and observe $w(1) = 1$.



Drift. The mean backward step is $-h$, toward the starting point.

Diffusion. The step variance is $h(1-h)$, so its leading term is still h.

For $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$, what are the **drift** and **diffusion** in reverse time?

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

- Here the score function appears: $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.
- Since $\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)$ (PF-ODE), the reverse drift equals $\mathbf{f} - g^2\mathbf{s} = \mathbf{v} - \frac{1}{2}g^2\mathbf{s}$.

Proof Sketch

- Convert the initial SDE to a probability flow ODE.
- Reverse the probability flow ODE.
- Convert the reversed probability flow ODE back to an SDE.

Anderson B. D. O. Reverse-time diffusion equation models, 1982
Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Proof of Reverse SDE

1. Convert the initial SDE to a probability flow ODE.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

2. Reverse the probability flow ODE by setting $\tau = 1 - t$, so $dt = -d\tau$.

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1 - \tau) + \frac{1}{2} g^2(1 - \tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$

3. To restore diffusion $g(1 - \tau)$, add the score term from step 2 once more to the drift.

$$d\mathbf{x} = \left(-\mathbf{f}(\mathbf{x}, 1 - \tau) + g^2(1 - \tau) \frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau + g(1 - \tau)\, d\mathbf{w}$$

   Here $\mathbf{w}(\tau)$ is a Wiener process in forward τ-time.

4. Return to $t = 1 - \tau$. Now $dt = -d\tau < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)\, d\bar{\mathbf{w}}.$$

Reverse SDE

Theorem

There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

$$d\mathbf{x} = \left(\mathbf{f}(\mathbf{x}, t) - g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\right) dt + g(t)d\bar{\mathbf{w}}$$

where $dt < 0$ and $\bar{\mathbf{w}}(t)$ is a reverse-time Wiener process.

Example · A one-dimensional Wiener process

Forward SDE	$dx_t = dw_t, \qquad x_0 = 0, \qquad f = 0, \quad g = 1.$
Reverse SDE	$dx_t = -s(x_t, t)\, dt + d\bar{w}_t, \qquad dt < 0.$
Gaussian score	$p_t = \mathcal{N}(0, t), \quad s(x, t) = \partial_x \log p_t(x) = -\dfrac{x}{t} \quad \Longrightarrow \quad dx_t = \dfrac{x_t}{t}\, dt + d\bar{w}_t.$
Euler–Maruyama step	$x_{t-h} \approx x_t - \dfrac{h x_t}{t} + \sqrt{h}\,\epsilon, \qquad 0 < h < t, \quad \epsilon \sim \mathcal{N}(0, 1).$

Starting at $x_1 = 1$, one backward step gives $x_{1-h} \approx 1 - h + \sqrt{h}\,\epsilon$.

Reverse SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

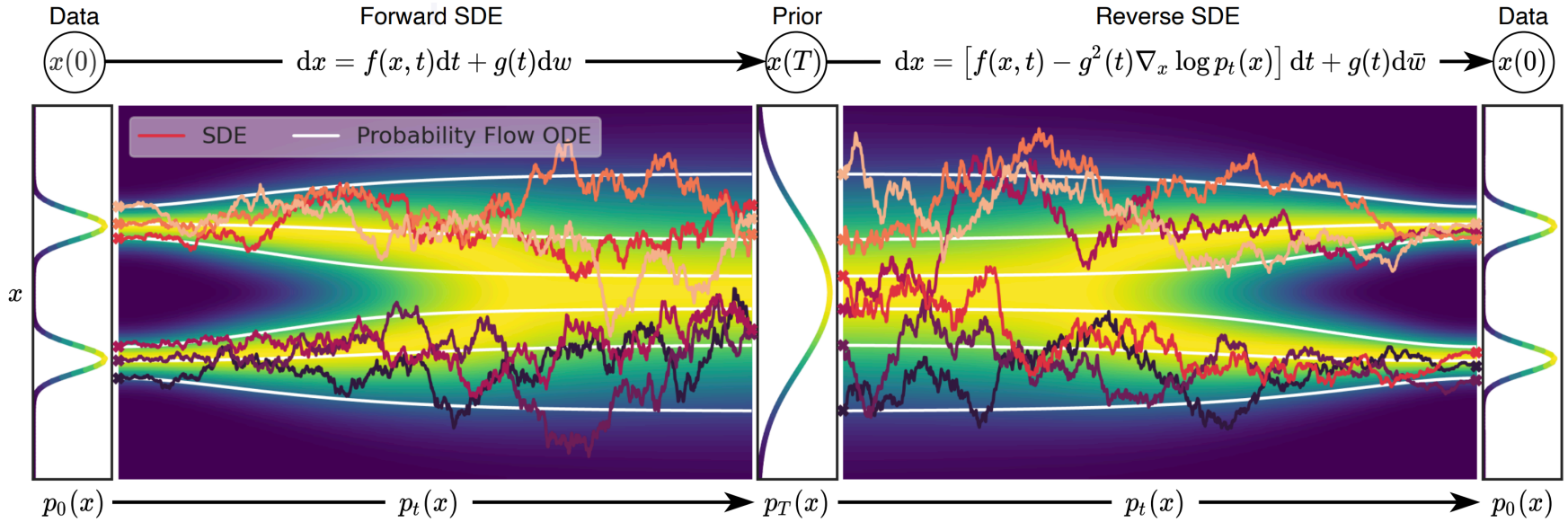
$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t), \quad \text{where} \quad \mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

Probability Flow ODE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt.$$

Reverse SDE

$$d\mathbf{x} = \left(\mathbf{v}(\mathbf{x}, t) - \tfrac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)\right) dt + g(t)\, d\bar{\mathbf{w}}.$$

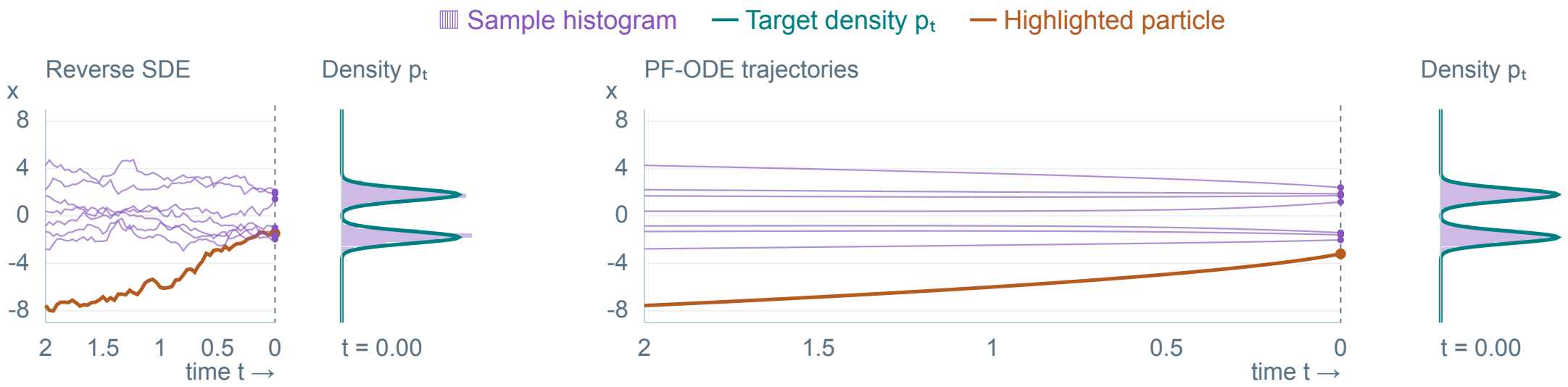


Reverse SDE and Probability Flow ODE

Start from the same noisy samples: two dynamics, the same evolving density.

$$\text{Reverse SDE:} \quad d\mathbf{x} = 2\mathbf{s}(\mathbf{x}, 2-\tau)d\tau + \sqrt{2}d\mathbf{w}, \quad t = 2 - \tau$$

From the same terminal samples at t = 2 back to t = 0.



Reverse SDE and PF-ODE follow the same densities back to data, with different particle trajectories.

Exact reverse conditional transitions; PF-ODE via quantiles. Reverse SDE does not retrace individual forward paths.

VE and VP: Probability Flow and Reverse SDE

Start from the forward SDE $d\mathbf{x} = \mathbf{f}\,dt + g\,d\mathbf{w}$ with score $\mathbf{s} = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$.

General formula	VE / NCSN	VP / DDPM
Forward coefficients Drift $\mathbf{f}(\mathbf{x}, t)$ and diffusion $g(t)$	$\mathbf{f} = 0, \qquad g^2 = \dfrac{d\sigma^2(t)}{dt}$ $\sigma(t)$ increases with time.	$\mathbf{f} = -\frac{1}{2}\beta(t)\mathbf{x}, \qquad g^2 = \beta(t)$ Unit initial variance is preserved.
Probability Flow ODE $d\mathbf{x} = (\mathbf{f} - \frac{1}{2}g^2\mathbf{s})\,dt$	$d\mathbf{x} = -\frac{1}{2}\dfrac{d\sigma^2(t)}{dt}\mathbf{s}\,dt$	$d\mathbf{x} = \left(-\frac{1}{2}\beta(t)\mathbf{x} - \frac{1}{2}\beta(t)\mathbf{s}\right)dt$
Reverse SDE $d\mathbf{x} = (\mathbf{f} - g^2\mathbf{s})\,dt + g\,d\bar{\mathbf{w}}$	$d\mathbf{x} = -\dfrac{d\sigma^2(t)}{dt}\mathbf{s}\,dt + \sqrt{\dfrac{d\sigma^2(t)}{dt}}\,d\bar{\mathbf{w}}$	$d\mathbf{x} = \left(-\frac{1}{2}\beta(t)\mathbf{x} - \beta(t)\mathbf{s}\right)dt + \sqrt{\beta(t)}\,d\bar{\mathbf{w}}$

For the reverse SDE, $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

Outline

01 Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)

02 Probability Flow ODE

03 Reverse SDE

04 Score-Based Generative Models Through SDEs

Score-Based Generative Models Through SDEs

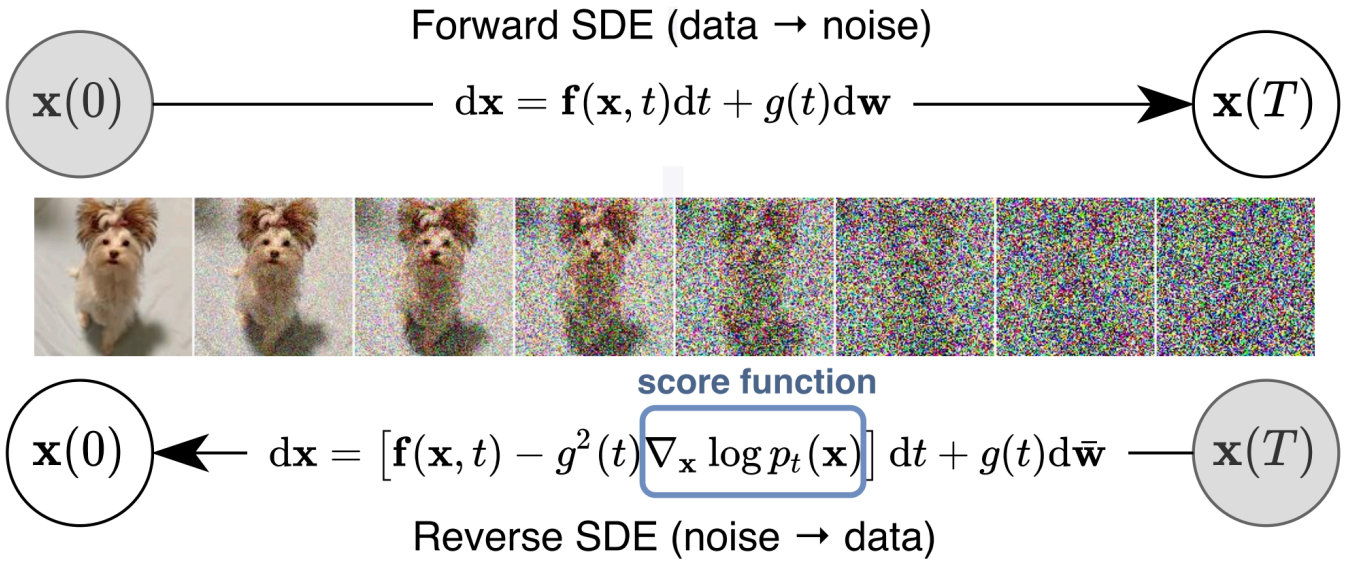
Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

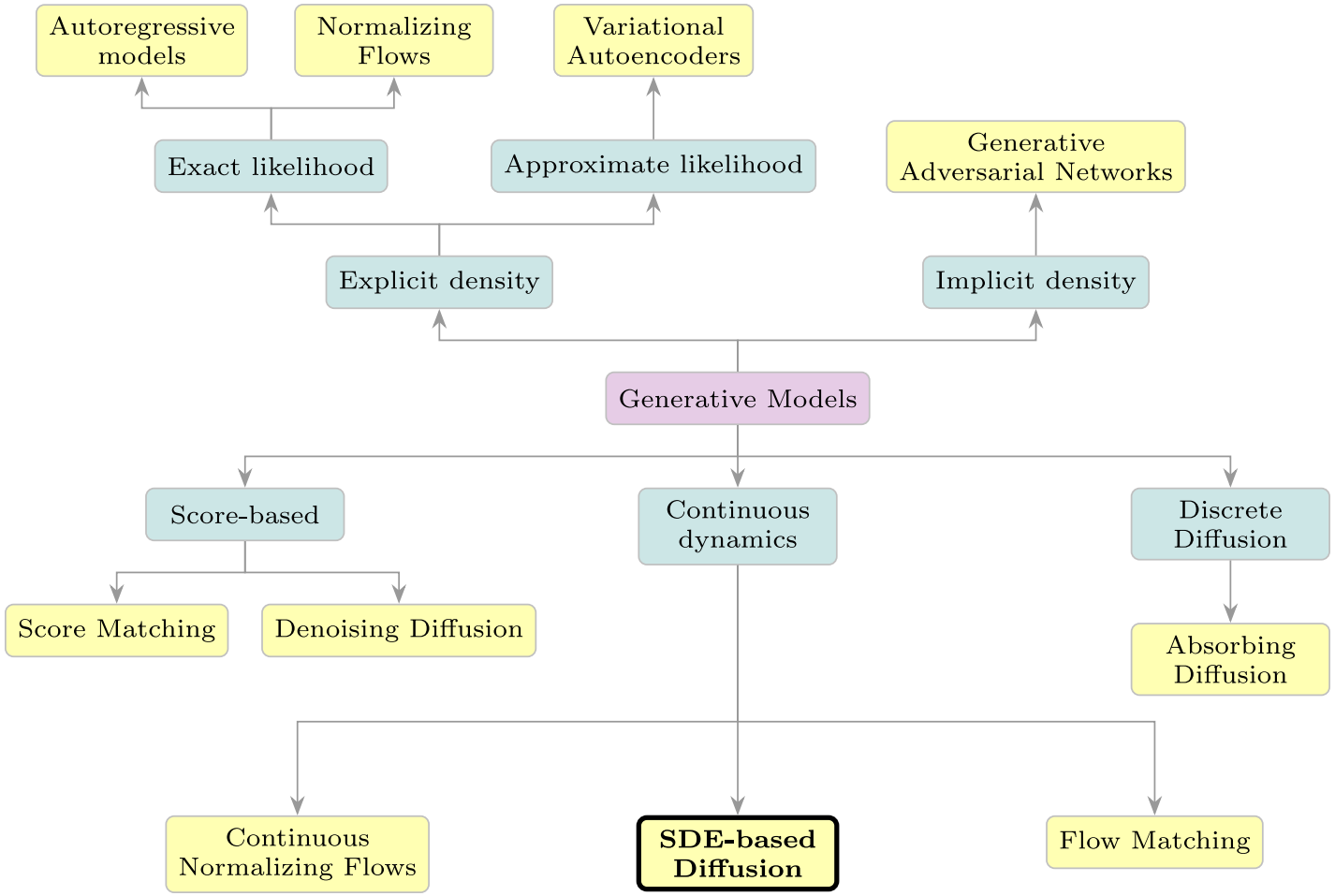
Use unit score-loss weights here. Can we extend this objective to continuous time?

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t) - \nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$



Generative Models Taxonomy



What Is the Score Target for VE and VP?

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t)-\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)\mid\mathbf{x}(0))\right\|_2^2$$

Fix a clean example $\mathbf{x}_0=\mathbf{x}(0)$. Each forward step uses fresh noise $\boldsymbol{\epsilon}_k\sim\mathcal{N}(0,\mathbf{I})$.

VE / NCSN

$$\mathbf{x}_k=\mathbf{x}_{k-1}+\sqrt{\sigma_k^2-\sigma_{k-1}^2}\,\boldsymbol{\epsilon}_k$$

VP / DDPM

$$\mathbf{x}_k=\sqrt{1-\beta_k}\,\mathbf{x}_{k-1}+\sqrt{\beta_k}\,\boldsymbol{\epsilon}_k$$

After two steps, the noise is a weighted sum of two independent Gaussian draws, so it is Gaussian again. Each further step only rescales it and adds fresh Gaussian noise.

In continuous time, the conditional law is still Gaussian. Write $\boldsymbol{\mu}=\boldsymbol{\mu}(\mathbf{x}(0),t)$ for its mean.

Conditional density

$$q(\mathbf{x}(t)\mid\mathbf{x}(0))=\mathcal{N}\big(\boldsymbol{\mu},\,\sigma^2(t)\mathbf{I}\big)$$

Score target

$$\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)\mid\mathbf{x}(0))=\frac{\boldsymbol{\mu}-\mathbf{x}(t)}{\sigma^2(t)}$$

How do $\boldsymbol{\mu}(\mathbf{x}(0),t)$ and $\sigma^2(t)$ evolve under the VE and VP forward SDEs?

Conditional Mean and Variance of the Forward SDE

With affine drift and additive Gaussian noise, the transition law is Gaussian.

$$d\mathbf{x} = f(t)\mathbf{x}\,dt + g(t)\,d\mathbf{w} \quad \Longrightarrow \quad q(\mathbf{x}(t) \mid \mathbf{x}(0)) = \mathcal{N}\big(\boldsymbol{\mu}(\mathbf{x}(0), t),\, \sigma^2(t)\mathbf{I}\big)\,.$$

Theorem

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t), \qquad \frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t).$$

Proof for the conditional mean

What happens to the Brownian increment when we average over paths starting at the same $\mathbf{x}(0)$?

$$\mathbb{E}[d\mathbf{x} \mid \mathbf{x}(0)] = f(t)\,\mathbb{E}[\mathbf{x}(t) \mid \mathbf{x}(0)]\,dt + g(t)\,\mathbb{E}[d\mathbf{w} \mid \mathbf{x}(0)].$$

The Brownian increment is independent of $\mathbf{x}(0)$ and has mean zero. Hence

$$d\boldsymbol{\mu}(\mathbf{x}(0), t) = \mathbb{E}[d\mathbf{x} \mid \mathbf{x}(0)] = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)\,dt \quad \Longrightarrow \quad \frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t).$$

Conditional Mean and Variance for VE and VP

Fix $\mathbf{x}(0)$ and write $v(t)$ for the conditional variance per coordinate. Then $\boldsymbol{\mu}(0) = \mathbf{x}(0)$ and $v(0) = 0$.

	VE / NCSN	VP / DDPM
SDE coefficients	$f(t) = 0, \quad g^2(t) = \dfrac{d\sigma^2(t)}{dt}$	$f(t) = -\frac{1}{2}\beta(t), \quad g^2(t) = \beta(t)$
Conditional mean $\dfrac{d\boldsymbol{\mu}}{dt} = f(t)\boldsymbol{\mu}(t)$	$\dfrac{d\boldsymbol{\mu}}{dt} = 0$ $\boldsymbol{\mu}(t) = \mathbf{x}(0)$	$\dfrac{d\boldsymbol{\mu}}{dt} = -\frac{1}{2}\beta(t)\boldsymbol{\mu}(t)$ $\boldsymbol{\mu}(t) = \exp\left(-\frac{1}{2}\int_0^t \beta(s)\,ds\right)\mathbf{x}(0)$
Conditional variance $\dfrac{dv}{dt} = 2f(t)v(t) + g^2(t)$	$\dfrac{dv}{dt} = \dfrac{d\sigma^2(t)}{dt}$ $v(t) = \sigma^2(t) - \sigma^2(0)$	$\dfrac{dv}{dt} = \beta(t)(1 - v(t))$ $v(t) = 1 - \exp\left(-\int_0^t \beta(s)\,ds\right)$

VE keeps the mean at $\mathbf{x}(0)$. **VP** shrinks the mean towards zero as variance grows towards one.

Score-Based Generative Models Through SDEs

Training

1. Sample $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$, $t \sim U[0, 1]$.
2. Sample $\mathbf{x}(t) \sim q(\mathbf{x}(t)|\mathbf{x}(0))$.
3. Compute loss $\mathcal{L} = \left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$.

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\Big(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\Big)$$

NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), \left[\sigma^2(t) - \sigma^2(0)\right] \cdot \mathbf{I}\right)$$

DDPM

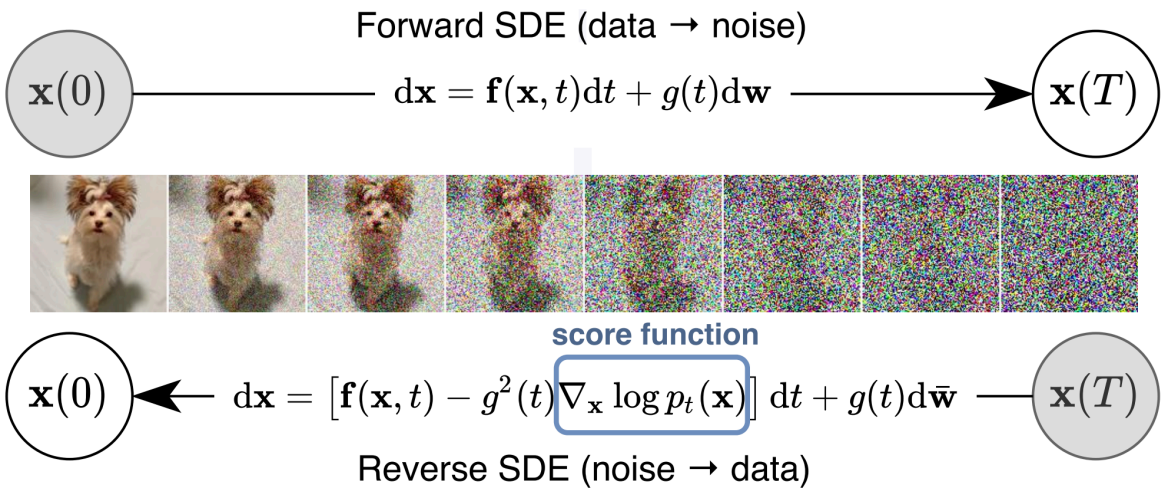
$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0) e^{-\frac{1}{2}\int_0^t \beta(s) ds}, \left(1 - e^{-\int_0^t \beta(s) ds}\right) \cdot \mathbf{I}\right)$$

Sampling with the Reverse SDE

Use the trained score for sampling.

Sampling

1. For the VP-SDE, sample $\mathbf{x}(1) \sim \mathcal{N}(0, \mathbf{I})$ from the approximate terminal distribution.
2. Solve the reverse SDE from $T = 1$ to $t = 0$ using `SDESolve`.



- An Euler–Maruyama step of the reverse SDE adds fresh Gaussian noise.
- An Euler step of the probability flow ODE is deterministic given the current state.

Sampling: Reverse SDE vs PF-ODE

Use the same VP prior sample $\mathbf{x}(1)$ and trained score $\mathbf{s}_{\boldsymbol{\theta}}$ for both methods.

Take K steps from $t = 1$ to $t = 0$ with $\Delta t = 1/K$.

Sampling (Reverse SDE, Euler–Maruyama)

1. Draw independent noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ at each step.
2. Update the state.

$$\mathbf{x}(t - \Delta t) = \mathbf{x}(t) - \Big[\mathbf{f}(\mathbf{x}(t), t) - g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\Big]\Delta t + g(t)\sqrt{\Delta t}\, \boldsymbol{\epsilon}.$$

Sampling (PF-ODE, Euler)

1. Update the state deterministically.

$$\mathbf{x}(t - \Delta t) = \mathbf{x}(t) - \Big[\mathbf{f}(\mathbf{x}(t), t) - \tfrac{1}{2}\, g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\Big]\Delta t.$$

Summary

- The VE-SDE matches the Gaussian perturbations used by NCSN. The VP-SDE is the continuous-time limit of the DDPM forward chain.

- The probability flow ODE follows the same density p_t as the diffusion. Its invertible flow gives the log-likelihood through the instantaneous change of variables.

- The reverse SDE uses the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ and draws fresh noise at each Euler–Maruyama step. The PF-ODE gives a deterministic trajectory from the same starting point.

- For VE and VP, the Gaussian transition gives an explicit conditional score target for continuous-time DSM.